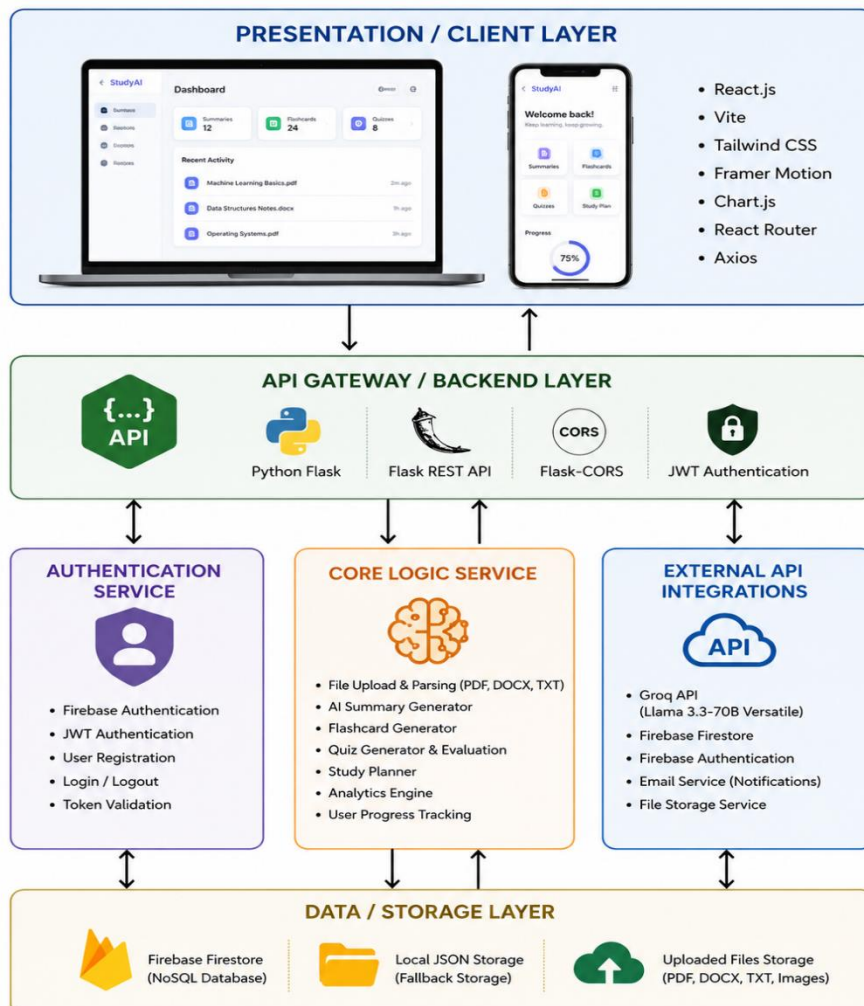


Solution Architecture

Date	11 July 2026
Team ID	XXXXXX
Project Name	StudyAI – Smart AI Study Companion & Quiz Generator
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table:

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for authentication, study material upload, AI summaries, flashcards, quizzes, study plans, analytics, and dashboard.	React.js, Vite, Tailwind CSS, Framer Motion, Chart.js, React Router, Axios
API Gateway	Receives requests from the frontend, validates authentication, routes API calls, handles CORS, and communicates with backend services.	Flask, Flask-CORS, JWT Authentication, REST APIs
Microservices	Processes uploaded files, communicates with the Groq AI model, generates summaries, flashcards, quizzes, study plans, manages analytics, and performs business logic.	Python Flask, Groq SDK (Llama 3.3-70B Versatile), PyPDF2, docx2txt
Database	Stores user accounts, uploaded study materials, AI-generated content, quiz results, study plans, and analytics with fallback storage support.	Firebase Firestore, Firebase Authentication, Local JSON Storage